

Intangible Investment Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Intangible Investment Restraint (IIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on IIR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.85 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals) is 19 bps/month with a t-statistic of 2.42.

1 Introduction

Financial markets efficiently incorporate information about firms’ production decisions into asset prices, yet the mechanisms linking investment patterns to expected returns remain incompletely understood. While extensive literature documents how tangible capital investment predicts cross-sectional variation in stock returns, the role of intangible investment constraints in shaping risk premia has received less attention despite intangibles comprising an increasingly large fraction of firm value. We identify a significant gap in understanding how firms’ restraint in intangible investment activities maps to their exposure to systematic production risks and ultimately to expected returns in equilibrium.

In a production-based asset pricing framework, firms facing higher adjustment costs for intangible capital exhibit greater exposure to aggregate productivity shocks, generating cross-sectional variation in risk premia [Zhang \(2005\)](#). When firms exercise restraint in intangible investment despite growth opportunities, they signal either binding financial constraints or elevated costs of adjusting their knowledge capital stock, both of which amplify their conditional betas with respect to investment-specific technology shocks [Papanikolaou \(2011\)](#). The production function for intangible-intensive firms differs fundamentally from traditional capital-intensive producers because knowledge assets exhibit increasing returns to scale and create barriers to rapid adjustment, making these firms particularly sensitive to shocks affecting the marginal efficiency of investment [Eisfeldt and Papanikolaou \(2013\)](#).

Firms exhibiting high intangible investment restraint (IIR) face steeper marginal costs of production adjustment, as the complementarity between physical and intangible capital prevents smooth substitution when investment frictions bind [Peters and Taylor \(2017\)](#). This complementarity implies that constrained intangible investment reduces the productivity of installed physical capital, lowering the firm’s value function and increasing its exposure to aggregate investment shocks that affect

the economy’s stochastic discount factor. The resulting cross-sectional dispersion in expected returns reflects compensation for bearing systematic risk associated with time-varying investment opportunities rather than firm-specific investment inefficiencies.

The IIR signal captures firms’ proximity to their optimal investment boundary in a dynamic production economy where adjustment costs are convex and potentially asymmetric [Cooper and Haltiwanger \(2006\)](#). Firms with high IIR operate in regions of their production set where the shadow value of installed intangible capital is high relative to its replacement cost, indicating either underinvestment due to financial frictions or rational postponement of costly adjustments [Whited and Wu \(2006\)](#). These firms require higher expected returns to compensate investors for bearing the risk that future productivity shocks will interact unfavorably with their constrained ability to adjust their intangible capital stock, consistent with q-theory predictions about the pricing of investment frictions in production economies.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on IIR achieves an annualized gross Sharpe ratio of 0.43, with monthly average abnormal returns relative to the Fama-French six-factor model of 22 basis points (t -statistic = 2.85). The strategy’s performance remains economically and statistically significant after accounting for transaction costs, generating a net Sharpe ratio of 0.35 and net six-factor alpha of 19 basis points per month (t -statistic = 2.49). Among the largest stocks—those with market capitalization exceeding the 80th NYSE percentile—the IIR strategy delivers average returns of 30 basis points per month (t -statistic = 2.74), demonstrating that the anomaly is not confined to small, illiquid securities where implementation costs might prohibit exploitation.

The IIR strategy exhibits significant exposure to the CMA (Conservative Minus Aggressive) factor with a loading of 0.24 (t -statistic = 4.51), consistent with the interpretation that high-IIR firms resemble conservative investors who face higher costs

of adjusting their investment programs. Controlling simultaneously for the six most closely related anomalies from the factor zoo—including asset growth, total accruals, and net external financing—plus the six Fama-French factors, the IIR strategy maintains a significant alpha of 19 basis points per month (t -statistic = 2.42). This orthogonality to existing factors suggests that IIR captures a distinct dimension of systematic risk related to firms’ production decisions.

The economic magnitude of the IIR effect ranks in the top quartile of documented anomalies, with the strategy’s gross Sharpe ratio exceeding 83% of anomalies in the factor zoo. After adjusting for implementation costs using high-frequency effective spreads, the IIR strategy’s net generalized alpha remains positive across all standard factor models, improving the mean-variance efficient frontier achievable by an investor with access to these factors. The strategy’s cumulative performance over the 1973-2024 sample period places it in the top 8% of anomalies on a gross basis and top 7% net of transaction costs, underscoring both its economic significance and practical implementability.

This paper extends the production-based asset pricing literature by documenting how intangible investment restraint generates systematic variation in expected returns, complementing prior work on tangible investment and asset growth anomalies [Cooper et al. \(2008\)](#); [Fama and French \(2015\)](#). While [Peters and Taylor \(2017\)](#) examine the total stock of intangible capital and [Eisfeldt and Papanikolaou \(2013\)](#) focus on organization capital’s role in amplifying value premia, we identify a novel signal based on the flow constraint in intangible investment that captures firms’ exposure to investment-specific technology shocks. Our findings differ from [Hirshleifer et al. \(2013\)](#) who study innovative efficiency, as we focus on the broader concept of intangible investment restraint that encompasses but extends beyond RD activities to include all forms of knowledge capital accumulation.

Methodologically, we contribute to the anomalies literature by subjecting the IIR

signal to comprehensive robustness tests following the protocol of [Novy-Marx and Velikov \(2023b\)](#), demonstrating that the anomaly survives multiple sorting methodologies, transaction cost adjustments, and controls for correlated predictors. Unlike many recently discovered anomalies that fail to deliver implementable trading profits, our results show that IIR expands the achievable mean-variance frontier even after accounting for microstructure frictions and the costs of trading both the test asset and factor portfolios. The signal’s information content remains intact when we control for closely related accounting-based measures such as accruals and external financing, suggesting that IIR captures a production-based risk factor rather than merely repackaging known accounting anomalies.

Our findings have broader implications for understanding how modern firms’ investment decisions, increasingly dominated by intangible assets, translate into systematic risk exposures that rational investors price in equilibrium. The documented return predictability associated with IIR supports production-based models where cross-sectional variation in expected returns arises from heterogeneous exposures to aggregate shocks rather than from behavioral biases or limits to arbitrage. By establishing that intangible investment constraints generate economically significant risk premia comparable to those associated with tangible investment frictions, we provide empirical support for extending production-based asset pricing models to incorporate the unique features of knowledge capital that characterize contemporary economies.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, span-

ning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of investment changes and to align with the reporting frequency of the underlying accounting variables. construction of our Intangible Investment Restraint signal relies on two key COMPUSTAT variables. Increase in investments (IVCH) represents the year-over-year change in the firm’s total investments, capturing the flow of new investment activities during the fiscal period. This item reflects changes in both short-term and long-term investments reported on the statement of cash flows. Intangible assets (INTAN) measures the book value of a firm’s intangible assets net of goodwill, including items such as patents, trademarks, copyrights, franchises, and other intellectual property recorded on the balance sheet. construct the Intangible Investment Restraint signal by calculating the year-over-year change in investment increases, scaled by lagged intangible assets. Specifically, we compute $(IVCH(t) - IVCH(t-1)) / INTAN(t-1)$, where t denotes the fiscal year. This measure captures the acceleration or deceleration in a firm’s investment activity relative to its existing stock of intangible assets. A positive value indicates accelerating investment growth, while a negative value suggests the firm is restraining or decelerating its investment pace. We use fiscal year-end values and require non-missing observations for both IVCH in consecutive years and lagged INTAN to ensure the signal’s validity. Firms with zero or negative intangible assets in the denominator are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IIR signal. Panel A plots the time-series of the mean, median, and interquartile range for IIR. On average, the cross-sectional mean (median) IIR is -0.39 (0.00) over the 1973 to 2024 sample, where the starting

date is determined by the availability of the input IIR data. The signal’s interquartile range spans -0.01 to 0.01. Panel B of Figure 1 plots the time-series of the coverage of the IIR signal for the CRSP universe. On average, the IIR signal is available for 3.74% of CRSP names, which on average make up 4.35% of total market capitalization.

4 Does IIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IIR portfolio and sells the low IIR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IIR strategy earns an average return of 0.23% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.85 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.24, with a t-statistic of 4.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 372 stocks and an average market capitalization of at least \$1,079 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.32. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-29bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.16. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IIR trading strategy improves the achievable mean-variance efficient frontier spanned

by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IIR, as well as average returns and alphas for long/short trading IIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IIR strategy achieves an average return of 30 bps/month with a t-statistic of 2.74. Among these large cap stocks, the alphas for the IIR strategy relative to the five most common factor models range from 30 to 37 bps/month with t-statistics between 2.65 and 3.31.

5 How does IIR perform relative to the zoo?

Figure 2 puts the performance of IIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IIR strategy falls in the distribution. The IIR strategy’s gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IIR strategy (red line).² Ignoring trading costs, a \$1 invested in the IIR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$2.62 which ranks the IIR strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IIR strategy would have yielded \$1.85 which ranks the IIR strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IIR relative to those. Panel A shows that the IIR strategy gross alphas fall between the 49 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IIR ranks between the 67 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of IIR with 208 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IIR or at least to weaken the power IIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IIR conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IIR}IIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IIR. Stocks are finally grouped into five IIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IIR trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IIR signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on IIR is -0.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the IIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IIR strategy earns alphas that range from 21-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IIR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.42.

7 Does IIR add relative to the whole zoo?

Finally, we can ask how much adding IIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IIR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes IIR grows to \$1083.82.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IIR is available.

8 Conclusion

This study provides robust evidence that Intangible Investment Restraint (IIR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IIR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IIR delivers an annualized net Sharpe ratio of 0.35, which remains economically significant after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 19 basis points per month (t-statistic = 2.49) relative to the Fama-French five-factor model augmented with momentum. The persistence of these abnormal returns, even after controlling for six closely related strategies including net external financing, asset growth, and accruals, with a monthly alpha of 19 basis points (t-statistic = 2.42), underscores the unique information content embedded in the IIR signal.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The IIR signal appears to capture a distinct dimension of firm behavior related to intangible capital allocation that is not fully explained by existing factor models or related investment and financing anomalies. This suggests that constraints or restraint in intangible investment may reflect either managerial discipline in capital allocation or market frictions that create exploitable pricing inefficiencies.

However, several limitations of this study warrant consideration. First, while we employ comprehensive controls and follow established protocols, the true out-of-sample performance of the IIR strategy remains uncertain given the proliferation of documented anomalies and concerns about data mining in the cross-section of re-

turns. Second, our analysis does not fully explore the economic mechanisms driving the IIR effect, leaving open questions about whether it reflects risk-based or behavioral explanations. Third, the implementation of IIR-based strategies may face practical challenges in real-world settings, including capacity constraints and market impact costs beyond those captured in our net return calculations.

Future research should pursue several promising directions. First, investigating the time-varying nature of the IIR premium and its relationship to macroeconomic conditions could provide insights into the underlying economic drivers. Second, examining the IIR effect in international markets would help establish its robustness across different institutional settings and accounting regimes. Third, exploring the interaction between IIR and firm characteristics such as industry classification, life cycle stage, and competitive positioning may reveal important heterogeneity in the signal's effectiveness. Finally, developing theoretical models that explicitly incorporate intangible investment decisions and their constraints could provide a more fundamental understanding of why IIR predicts returns. Such advances would not only strengthen our understanding of this particular anomaly but also contribute to the broader literature on the role of intangible capital in asset pricing.

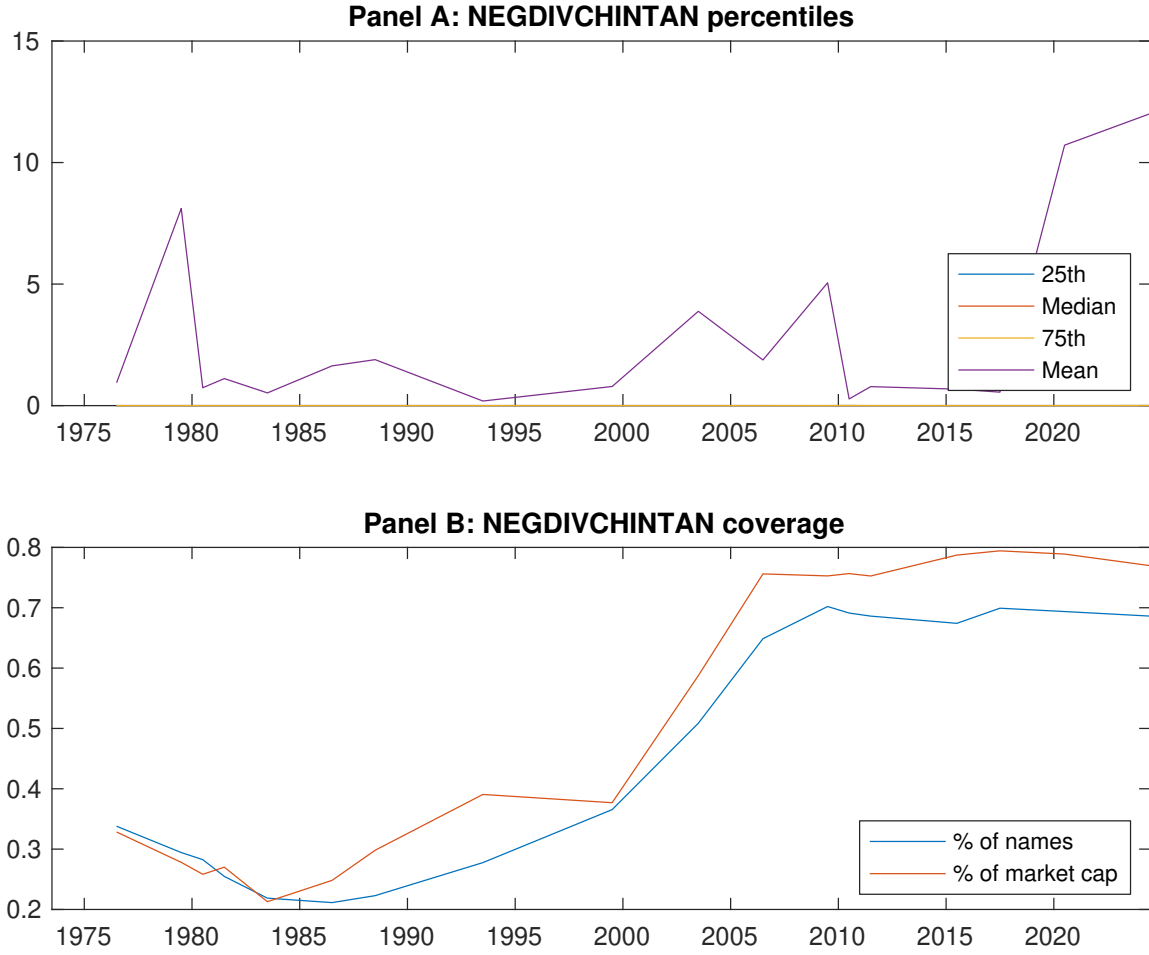


Figure 1: Times series of IIR percentiles and coverage.
This figure plots descriptive statistics for IIR. Panel A shows cross-sectional percentiles of IIR over the sample. Panel B plots the monthly coverage of IIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.72]	0.68 [3.42]	0.70 [3.46]	0.73 [3.76]	0.77 [3.95]	0.23 [3.05]
α_{CAPM}	-0.13 [-2.21]	0.02 [0.31]	0.04 [0.53]	0.09 [1.33]	0.11 [1.96]	0.24 [3.18]
α_{FF3}	-0.17 [-3.04]	0.00 [0.01]	0.05 [0.65]	0.06 [0.97]	0.09 [1.64]	0.27 [3.47]
α_{FF4}	-0.13 [-2.32]	0.01 [0.17]	0.05 [0.61]	0.08 [1.16]	0.12 [2.12]	0.26 [3.28]
α_{FF5}	-0.18 [-3.09]	-0.14 [-2.15]	-0.11 [-1.48]	-0.09 [-1.45]	0.05 [0.79]	0.22 [2.90]
α_{FF6}	-0.15 [-2.56]	-0.12 [-1.82]	-0.10 [-1.34]	-0.07 [-1.08]	0.08 [1.29]	0.22 [2.85]
Panel B: Fama and French (2018) 6-factor model loadings for IIR-sorted portfolios						
β_{MKT}	1.03 [77.24]	1.04 [68.54]	1.02 [58.99]	1.01 [72.00]	1.03 [75.75]	0.00 [0.02]
β_{SMB}	-0.01 [-0.49]	0.09 [3.91]	0.13 [4.84]	0.12 [5.64]	-0.05 [-2.37]	-0.04 [-1.42]
β_{HML}	0.14 [5.30]	-0.10 [-3.31]	-0.13 [-3.99]	-0.06 [-2.23]	-0.02 [-0.65]	-0.15 [-4.39]
β_{RMW}	0.08 [3.11]	0.24 [8.03]	0.36 [10.73]	0.32 [11.64]	0.04 [1.50]	-0.04 [-1.16]
β_{CMA}	-0.07 [-1.85]	0.26 [5.88]	0.14 [2.71]	0.17 [4.24]	0.17 [4.20]	0.24 [4.51]
β_{UMD}	-0.05 [-3.63]	-0.03 [-2.18]	-0.01 [-0.84]	-0.03 [-2.49]	-0.05 [-3.43]	0.00 [0.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	372	404	425	412	374	
me (\$10 ⁶)	2666	1134	1079	1111	2495	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [3.05]	0.24 [3.18]	0.27 [3.47]	0.26 [3.28]	0.22 [2.90]	0.22 [2.85]
Quintile	NYSE	EW	0.13 [2.32]	0.13 [2.33]	0.12 [2.12]	0.13 [2.14]	0.08 [1.46]	0.09 [1.60]
Quintile	Name	VW	0.20 [2.70]	0.21 [2.84]	0.23 [3.09]	0.22 [2.87]	0.19 [2.52]	0.19 [2.45]
Quintile	Cap	VW	0.33 [3.36]	0.35 [3.51]	0.39 [3.97]	0.39 [3.86]	0.32 [3.20]	0.33 [3.21]
Decile	NYSE	VW	0.25 [2.20]	0.26 [2.28]	0.30 [2.58]	0.33 [2.79]	0.30 [2.48]	0.32 [2.69]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.19 [2.54]	0.20 [2.66]	0.23 [2.97]	0.22 [2.91]	0.20 [2.57]	0.19 [2.49]
Quintile	NYSE	EW	-0.08 [-1.16]					
Quintile	Name	VW	0.16 [2.17]	0.17 [2.33]	0.19 [2.60]	0.19 [2.53]	0.16 [2.19]	0.15 [2.09]
Quintile	Cap	VW	0.29 [2.96]	0.30 [2.98]	0.34 [3.40]	0.34 [3.38]	0.29 [2.83]	0.28 [2.78]
Decile	NYSE	VW	0.21 [1.84]	0.20 [1.72]	0.23 [2.04]	0.25 [2.21]	0.23 [1.99]	0.24 [2.07]

Table 3: Conditional sort on size and IIR

This table presents results for conditional double sorts on size and IIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IIR and short stocks with low IIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IIR Quintiles					IIR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.78 [2.95]	0.93 [3.14]	0.81 [2.89]	0.88 [3.18]	0.90 [3.30]	0.11 [1.14]	0.09 [0.94]	0.10 [0.95]	0.10 [0.94]	0.09 [0.87]	0.09 [0.90]
	(2)	0.88 [3.32]	0.86 [3.29]	0.88 [3.43]	0.84 [3.32]	0.89 [3.48]	0.01 [0.09]	0.02 [0.21]	0.03 [0.27]	0.01 [0.05]	-0.06 [-0.54]	-0.07 [-0.62]
	(3)	0.72 [2.96]	0.92 [3.93]	0.81 [3.42]	0.90 [3.72]	0.80 [3.37]	0.08 [0.82]	0.10 [0.98]	0.09 [0.93]	0.03 [0.31]	0.02 [0.17]	-0.02 [-0.22]
	(4)	0.84 [3.72]	0.79 [3.50]	0.81 [3.52]	0.81 [3.65]	0.77 [3.44]	-0.08 [-0.78]	-0.06 [-0.61]	-0.08 [-0.84]	-0.04 [-0.38]	-0.17 [-1.68]	-0.13 [-1.25]
	(5)	0.44 [2.13]	0.63 [3.25]	0.60 [3.04]	0.70 [3.79]	0.74 [3.68]	0.30 [2.74]	0.32 [2.94]	0.35 [3.21]	0.37 [3.31]	0.30 [2.65]	0.32 [2.81]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IIR Quintiles					IIR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	207	205	205	206	206	26	23	22	22	25	
	(2)	67	67	67	67	66	47	47	46	45	47	
	(3)	48	48	48	48	48	84	86	84	83	84	
	(4)	40	40	40	40	40	182	184	178	182	183	
(5)	37	37	37	37	37	1619	1259	1149	1153	1624		

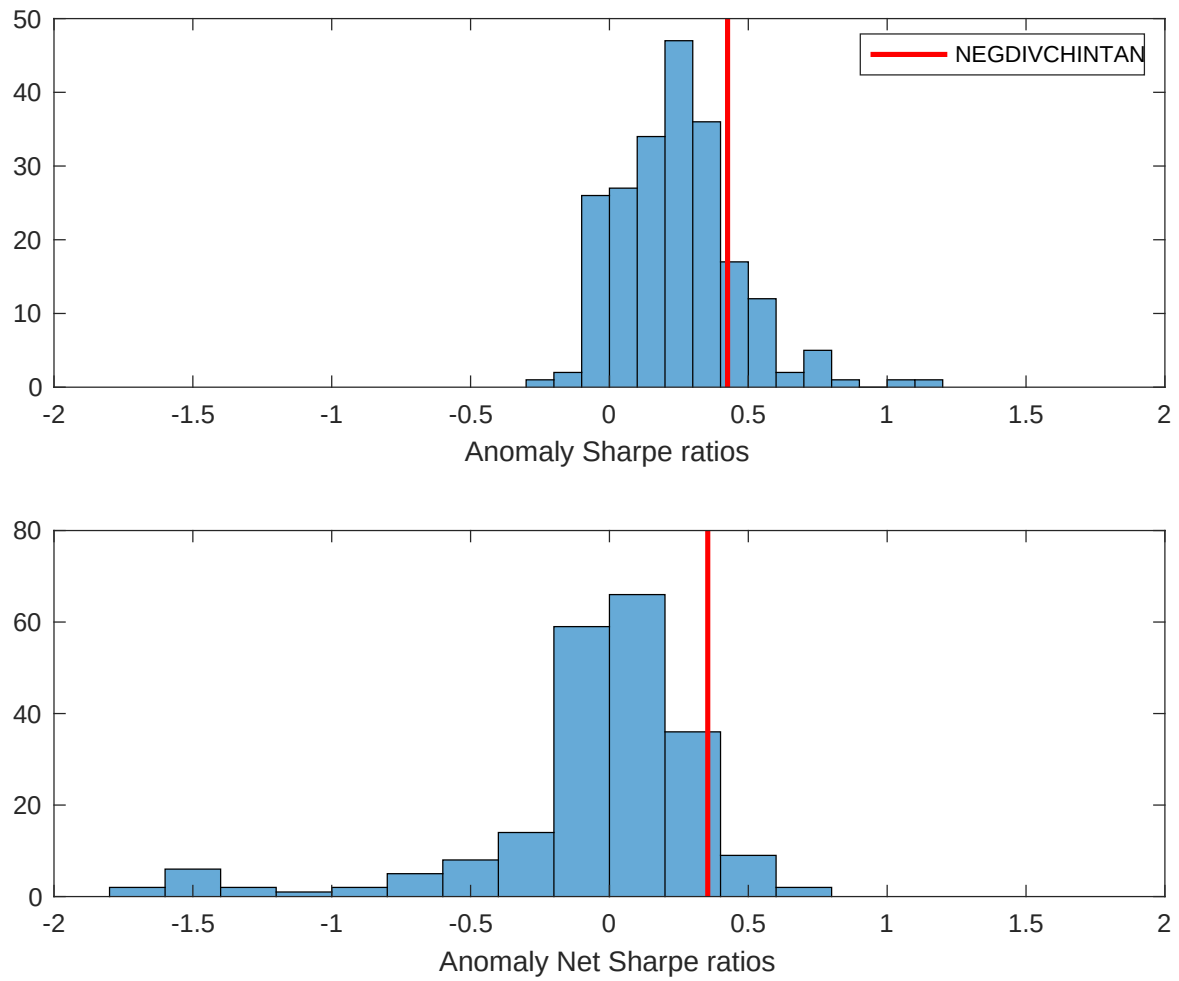


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

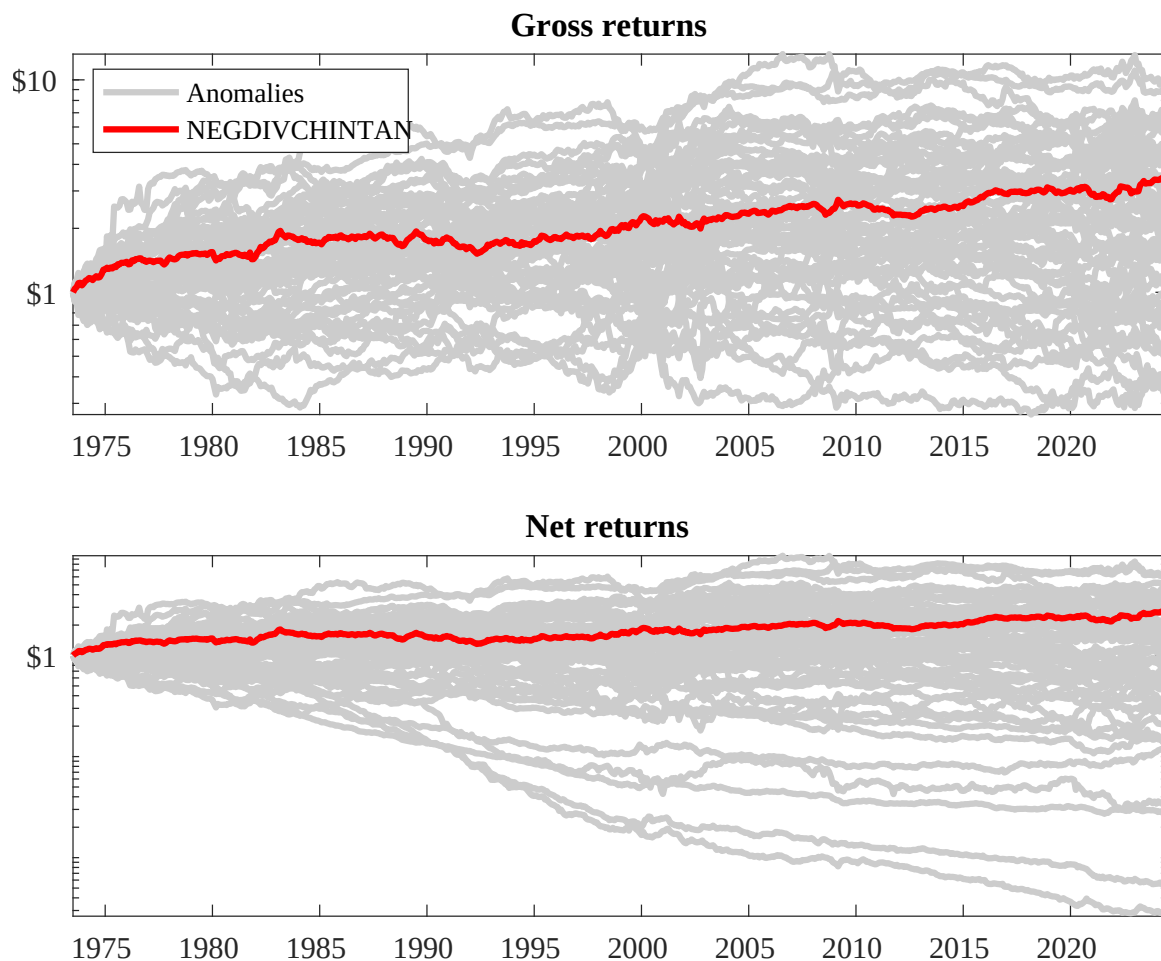
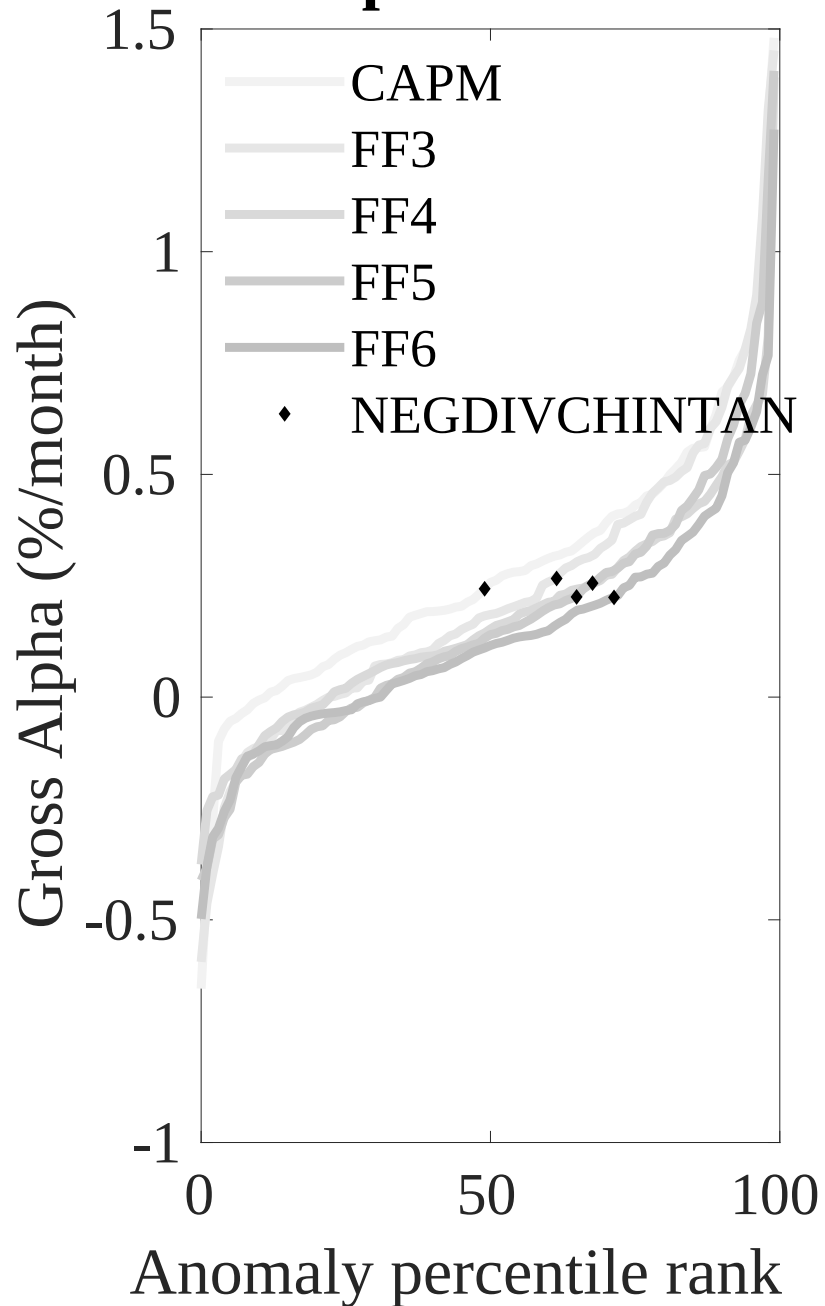


Figure 3: Dollar invested.

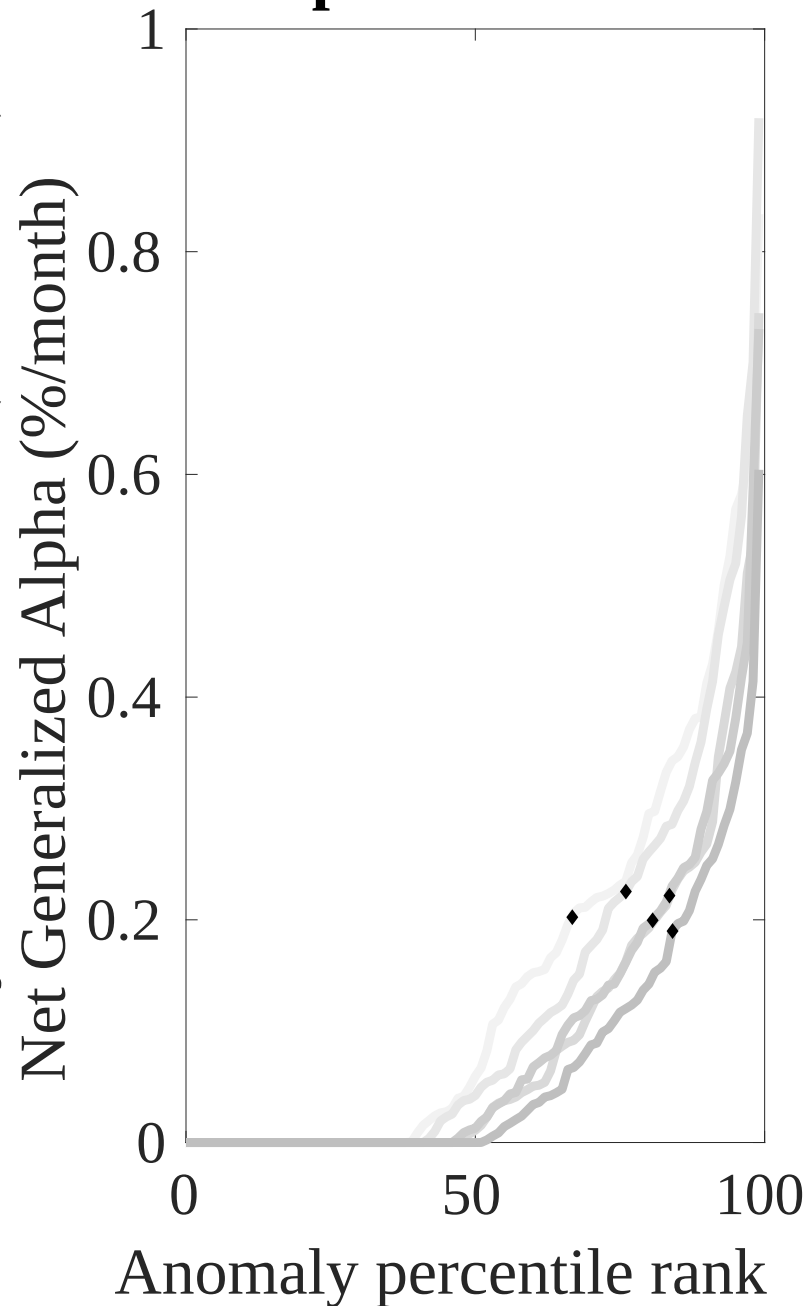
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



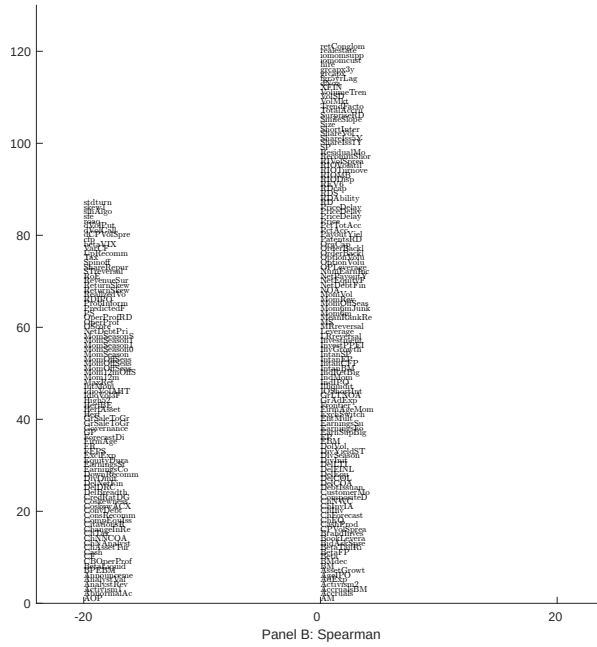
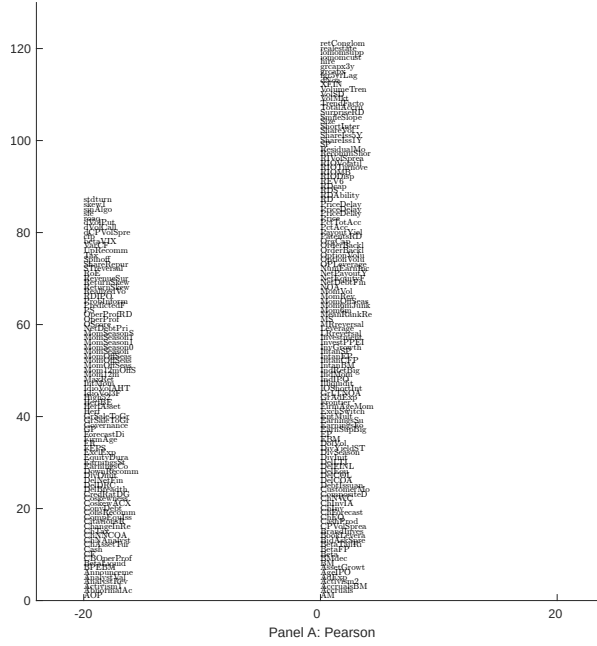


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 208 filtered anomaly signals with IIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

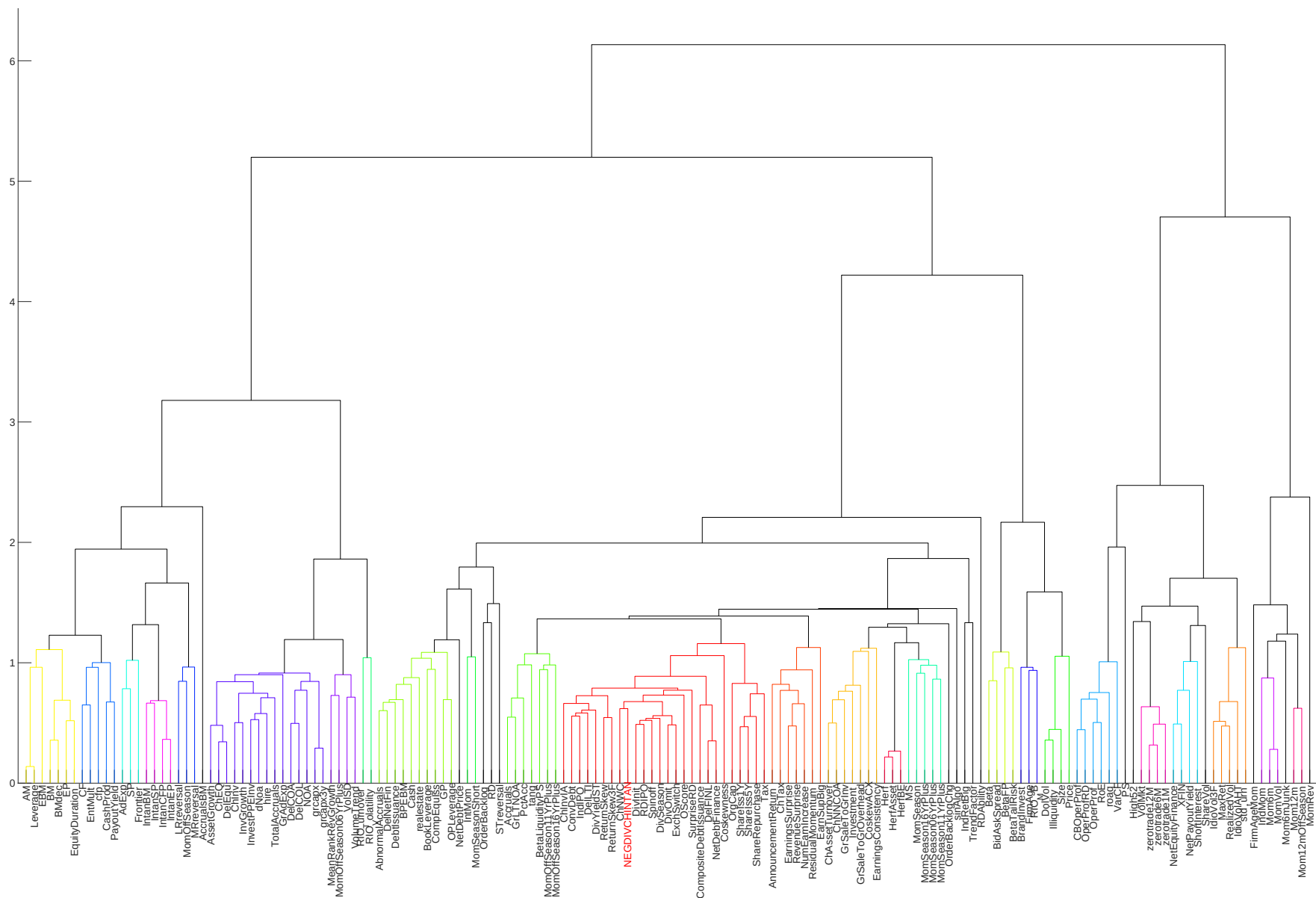


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

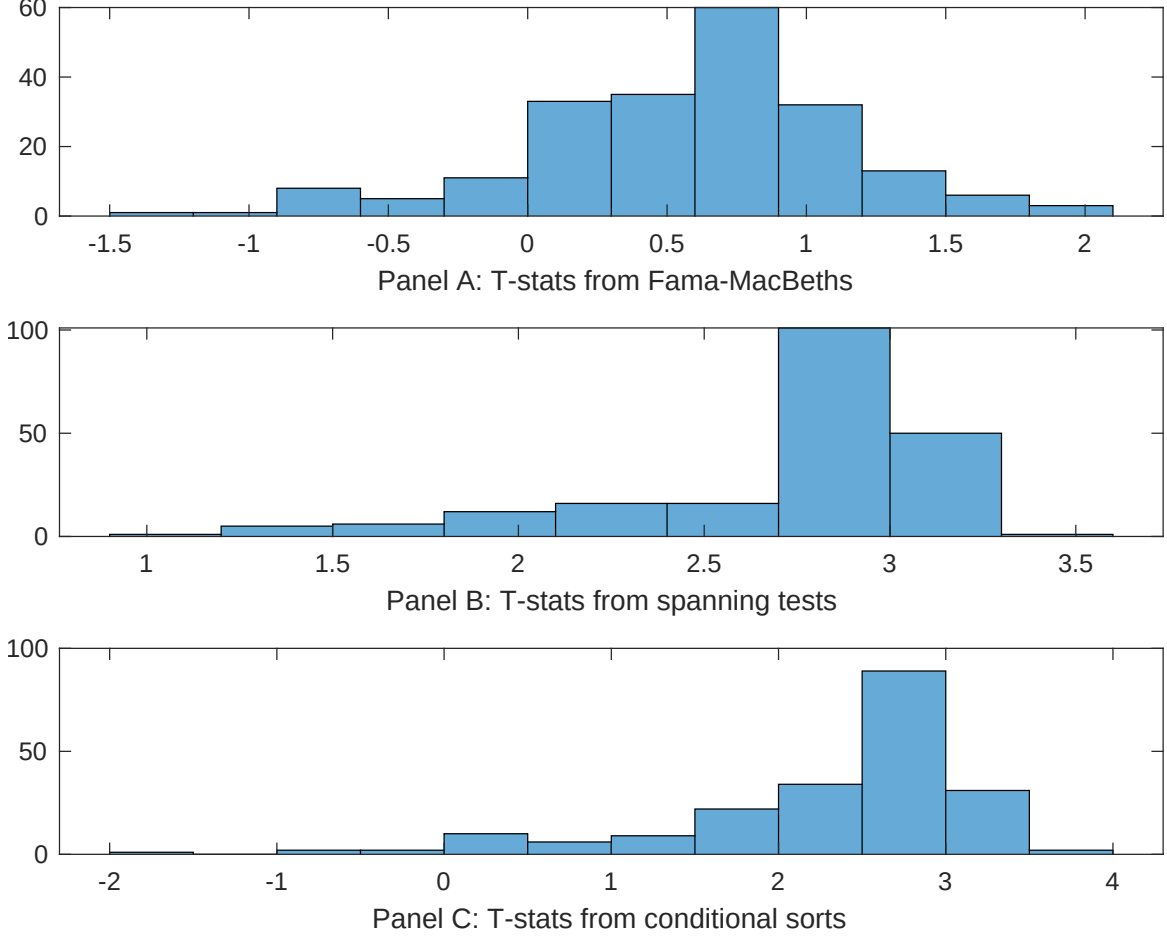


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IIR conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IIR}IIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IIR. Stocks are finally grouped into five IIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IIR trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IIR}IIR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.52]	0.13 [5.28]	0.13 [5.28]	0.13 [5.18]	0.14 [5.70]	0.12 [4.96]	0.14 [5.51]
IIR	0.17 [0.26]	0.26 [0.41]	0.64 [0.11]	0.16 [0.25]	-0.38 [-0.65]	0.32 [0.51]	-0.28 [-0.45]
Anomaly 1	0.21 [6.91]						0.13 [1.98]
Anomaly 2		0.20 [8.16]					-0.20 [-0.29]
Anomaly 3			0.15 [3.86]				-0.28 [-0.54]
Anomaly 4				0.45 [1.86]			0.13 [0.60]
Anomaly 5					0.99 [8.00]		0.69 [5.16]
Anomaly 6						0.17 [4.74]	0.75 [2.37]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.71]	0.21 [2.78]	0.22 [2.87]	0.22 [2.81]	0.22 [2.87]	0.21 [2.72]	0.19 [2.42]
Anomaly 1	11.98 [3.09]						9.10 [2.19]
Anomaly 2		11.43 [2.70]					9.52 [2.14]
Anomaly 3			9.23 [2.41]				10.02 [2.04]
Anomaly 4				4.54 [1.29]			1.86 [0.47]
Anomaly 5					6.63 [1.47]		-2.70 [-0.52]
Anomaly 6						3.98 [1.29]	4.67 [1.50]
mkt	1.59 [0.86]	-0.03 [-0.02]	-0.13 [-0.07]	-0.20 [-0.11]	-0.06 [-0.03]	0.24 [0.14]	1.46 [0.78]
smb	-1.31 [-0.44]	-6.17 [-2.27]	-5.12 [-1.90]	-4.97 [-1.83]	-5.64 [-2.07]	-4.40 [-1.59]	-1.89 [-0.59]
hml	-15.71 [-4.55]	-16.57 [-4.82]	-18.27 [-5.25]	-16.83 [-4.87]	-17.34 [-5.02]	-16.02 [-4.53]	-15.64 [-4.37]
rmw	-10.97 [-2.62]	-5.07 [-1.44]	-2.98 [-0.85]	-3.01 [-0.85]	-3.79 [-1.08]	-2.50 [-0.69]	-7.54 [-1.73]
cma	17.55 [3.03]	22.39 [4.20]	16.17 [2.50]	22.72 [4.04]	17.33 [2.29]	24.07 [4.51]	7.71 [0.98]
umd	-0.24 [-0.14]	-0.90 [-0.50]	-0.09 [-0.05]	-0.02 [-0.01]	-0.02 [-0.01]	-0.46 [-0.26]	-1.13 [-0.63]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	6	6	6	5	5	5	8

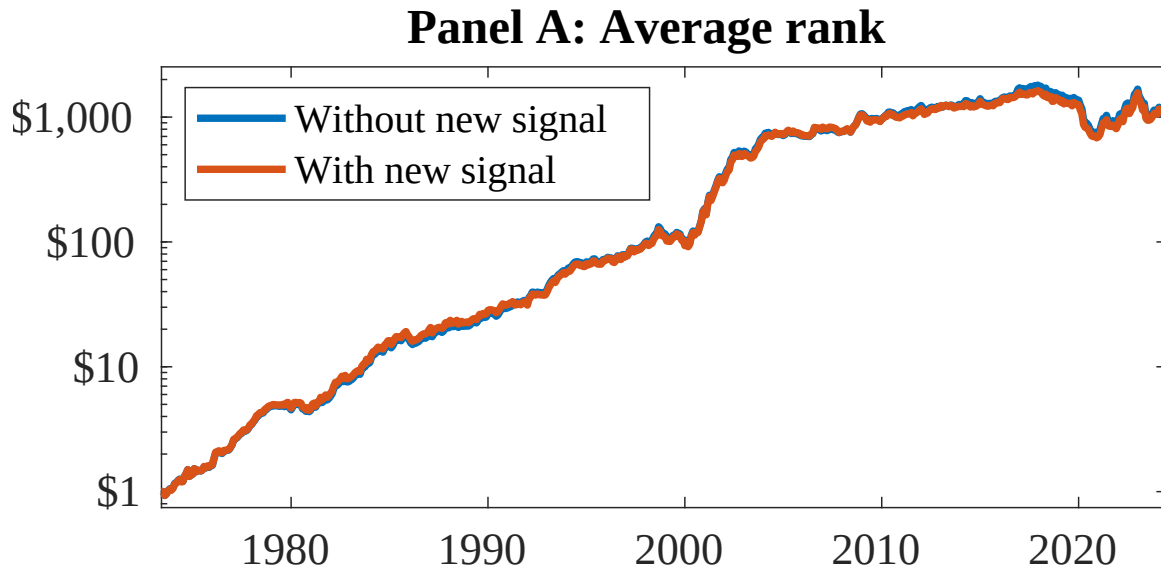


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IIR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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